

AI Shockwaves: What Higher Education CIOs Must Do Now

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Initiatives: Technologies and Markets

The evolving impacts of AI and changing public sentiment about the perceived value of higher education are combining to create shockwaves in the sector. This research reviews the opportunities for higher education CIOs to adapt technologies strategies to support and align with these challenges.

Insights at a Glance

Key Insights:

- Demand for curriculum agility and flexible credentialing is increasing, requiring CIOs to evaluate key technologies to enable lifelong learning and respond to changing workforce requirements.
- While those institutions that do not update strategies and embrace AI risk obsolescence, those that proactively adapt and digitally transform have opportunities to both survive and thrive.
- CIOs must help their institutions understand how they need to adapt to AI literacy across curricula and form strategic partnerships — both across the institution, across higher education and with key vendors.

Actionable Recommendations:

- **Assess maturity:** Evaluate current institutional AI maturity and AI readiness. The CIO is uniquely positioned to provide insight into the data readiness to support AI.
- **Reset AI ambition and governance:** Align AI investments and governance with your institution's mission, distinctive position and strategy, prioritizing use cases and tool selection.

- **Cultivate AI literacy and ethics:** Promote responsible AI use among faculty, staff and students, focusing on effective practice and targeting curriculum agility.
- **Optimize IT costs:** Strategically review IT costs and focus any resultant available IT team capacity toward building new AI-enabled capabilities.
- **Build communities of practice:** Accelerate adoption of AI-enabled teaching, research and administration by sharing knowledge of good practice and collaboration via use-case specific discussions.
- **Commit to continuous review:** Regularly refine education technology strategies and practices to drive appropriate ongoing institutional change in a time when AI disruption is accelerating. ¹

Issue Context

By 2030, most teaching, research and student content is projected to be AI-supported, leading to increased demands on institutional IT and increased partnerships among institutions, technology vendors and industry (see [The Future of Higher Education – Vision 2035](#)). The rapid emergence of AI in the higher education sector is generating shockwaves that demand action from higher education CIOs to support institutional success. The disruptive potential of AI applications across teaching, research and administration has now combined with broader drivers of higher education sector change. These include financial fragility, demographic shifts, tighter migration policies, evolving student expectations and decreasing revenue streams from international students.

These factors have led to institutional strategy resets around the world, with leadership teams seeking to reposition to survive and thrive in the face of intense global competition for students and research partnerships. Within these strategies, the need to both research AI and to ensure that universities deliver value in a world of AI have become key challenges.

For the CIO, there is a need to review and accelerate institutional AI practices that deliver an optimal path to appropriate AI adoption. Within higher education, AI adoption is accelerating but uneven. Early-stage pilots have typically focused on personal productivity, virtual assistants for administrative and teaching applications, and research enablement. However, generative AI (GenAI) slid into the Trough of Disillusionment on the Gartner Higher Education Hype Cycle in 2025 due to the frequent failure of these early-stage pilots to deliver tangible return on investment.

Many institutions remain AI-cautious. However, the sector has also seen the growth of both enterprise-grade AI solutions and GenAI embedded within education-specific products, such as the learning management system (LMS), CRM and ERP. GenAI use by students is now widespread, and institutional faculty and staff adoption is also building.

Higher education CIOs must align technology and AI strategies to institutional strategies that are adapting to address declining public confidence in existing education models and the larger secondary and tertiary impacts of AI (see [Walking the Golden Path to Value: 2025 IT Symposium/Xpo Keynote Insights](#)). A growing share of Americans (70%) now believe that higher education is going in the wrong direction, according to a recent Pew Research Center analysis. Only 28% say it's on track for success. Momentum is already building in many parts of the sector, particularly among institutions seeking to accelerate technology-enabled experimentation and innovation. Higher education CIOs are in a unique position to leverage their AI knowledge to help their institutions evolve how AI is incorporated into teaching and learning. Higher education CIOs must therefore review, refine and revolutionize technology capabilities to align with goals for new institutional value propositions in an AI-enabled world (see [The Future of Higher Education – Vision 2035](#)).

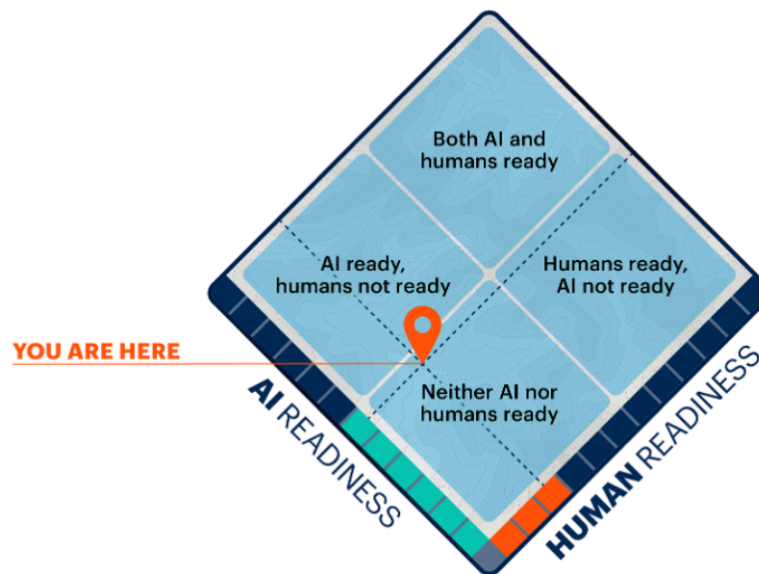
Impact Brief

- Within the global higher education sector, challenges to institutional reputations, enrollments and, in some cases, survival are impacting technology priorities. Public confidence in the value of higher education is divided, costs are rising ² and the perceived value of degrees is being questioned. ³ Institutional closures ⁴ and mergers are rising, and global student mobility patterns are changing. ⁵ A tougher market for entry-level graduate roles is emerging — both due to AI and broader economic environment shockwaves. ⁶ AI-enabled differentiation in value, experience and outcomes is becoming both a defensive necessity and a potential opportunity.
- Student use of AI is extensive and frequently outstripping that of faculty. ⁷ However, student expectations of graduating with AI-ready skills for employability demand that some elements of AI literacy become embedded across the curriculum. AI maturity is typically low, and refining institutional AI strategy, governance and literacy is now a priority.
- Governments around the world are promoting strong skills-based agendas. ⁸ Alternative credentialing — including badges, certificates, licenses, certifications and apprenticeships — is increasing. ⁹ Learning technology platforms must adapt and evolve to address a flexible portfolio of products that support these expanding learner needs for lifelong learning, reskilling and upskilling.
- As AI accelerates skill churn and narrows the half-life of technical skills, the path to becoming a world-class human performer becomes complex, impacting both human knowledge development needs and institutional ability to innovate. This acceleration of knowledge decay creates an urgent need for CIOs to empower faculty to continually reinvent the curriculum or risk development of teaching assets that may become obsolete by graduation.
- Policy and scrutiny are emerging: Regulators and national bodies are recognizing the need for appropriate AI adoption, with UNESCO urging human-centered frameworks, ¹⁰ the U.S. Department of Education permitting AI uses ¹¹ (tutoring, materials, advising) with responsible-use guardrails, and the EU AI Act encouraging appropriate ethical adoption by educators. ¹²

In preparation for continued impacts of these shockwaves, higher education CIOs must evolve both AI readiness and human AI readiness to enable competitiveness for an AI-ready world (see Figure 1). CIOs should seek opportunities to propel these two important priorities: adapting how students learn and supporting curriculum changes that enable graduates to compete in a world with AI.

Figure 1: The “You Are Here” Gartner Positioning System

The “You Are Here” Gartner Positioning System



Source: Gartner
01-3717

Gartner

Implications

The Call to Action

Higher education CIOs have two distinct roles in responding to the AI shockwaves: supporting student learning with AI and enabling new curriculum elements and teaching changes with AI. The CIO must design new digital capabilities to empower institutions facing AI shockwaves and multiple challenges of change:

- **Customer and stakeholder needs are evolving.** How must technology ecosystems support institutional needs for research, teaching, degrees and flexible credentials to support the needs of learners and stakeholders in an AI era?
- **Commercial models are changing.** How can higher education technologies deliver operational efficiencies while realigning to new business and operating models?
- **Culture and capacity of IT teams can be a barrier.** How can IT teams build appropriate capacity and capabilities to manage rapidly evolving demands?

- **Collaboration is needed.** What partnerships internally, across the sector, with AI vendors and with corporate employers can help drive appropriate change?
- **Curriculum agility is needed.** How can technology enable the creation and delivery of a more adaptable curriculum that embraces the latest employer needs and research insights?

Global government skills-based agenda mandates are creating a strong push toward job-based university models with emerging digital technologies such as AI, and they facilitate citizen development initiatives to meet the needs of an increasingly digital world.

However, AI's ripple effects on jobs, organizations and education needs are far from clear; the value of the critical thinking and judgment nurtured in a classic or jobs-based university will intensify as AI adoption builds (see [AI's Ripple Effect on Jobs and Organizations](#)).

Higher education institutions are therefore seeking to design future strategies that balance the tension between meeting urgent corporate skills-based training needs and cultivating higher-order student critical thinking, judgment, ethics and decision-making capabilities. Blending degrees and credentials ^{1 3} is likely as institutions prepare for the future, but the implications for technology and the CIO are clear: The platforms that have supported education to date need review and rethinking to support future target positions.

By 2030, institutions will seek a distinctive niche to deliver value in the market and adapt in response to continued market shockwaves. Universities will seek to identify a distinctive position that enables differentiation, and the higher education CIO must then enable that position through appropriate selection and evolution of AI and other technologies. AI will impact what institutions teach and how they teach.

This is not a time to confuse experimentation with strategy, history with complacency, or past business models as timeless for the future of education. As the education industry strives to rethink how best to prepare humans to face society's future challenges, CIOs must reshape an appropriate response to AI through deliberate review, reflection, collaboration and action.

What Not to Do Now

Higher education CIOs must work with their leadership teams to ensure that their institution:

- Does not confuse AI experimentation with long-term strategy
- Does not accept faculty resistance to change as an acceptable reason for delay; nor confuse uncoordinated faculty AI exploration with strategy
- Does not ignore opportunities for strategic cost optimization to target available resources toward AI and broader innovation needs

What Higher Education CIOs Must Do Now

Identify and communicate your ambition:

- Work with your leadership team to shape your distinctive position in an AI-enabled world (see [The Critical Relationships Higher Education CIOs Must Master to Expand Their Role](#)).
- Evaluate your current state of technology and identify the changes required to support your desired future state.
- Cultivate board-level understanding of the threats and opportunities of AI and alternative credentials to your institutional strategy.
- Manage the risks that skills-based credentialing needs and ongoing AI shockwaves might create for each element of your strategy.

Evolve your AI strategy:

- To enable new modes of education delivery, identify the current state of critical technology and AI solutions that will enable your target institutional business model (see [Roadmap to Accelerate AI Strategy for Higher Education CIOs](#) and [AI Maturity Assessment](#)).

- Take inspiration from institutions with leading AI visions; your AI strategy is the first step in developing a comprehensive AI roadmap:
 - “Students will gain skills, guided by personalized AI experiences, ensuring they graduate not just job-ready, but future-ready.” —Seneca Polytechnic ^{1 4}
 - “As organizations worldwide grapple with new regulatory obligations and ethical concerns, our mission is to provide innovative AI research and practical solutions.” —The Wharton School, The University of Pennsylvania ^{1 5}
 - “Develop next-generation AI systems through a variety of endeavors, ranging from fundamental research, educational activities, and partnerships with industry.” —National University of Singapore AI Laboratory. ¹⁶

Evaluate what is needed to accelerate and bring your institutional AI strategy to life:

- Reset institutional AI ambition, governance and investments.
- Target human AI readiness by building AI literacy, cultivating communities in key use-case areas and ensuring evolution of effective practices.

- Compare your organization with peers and leaders:
 - The University of Sydney boosted AI acceleration by enabling educators to build AI “agents” to answer questions, provide feedback, give advice, explain concepts, do role plays and guide learning activities. Educators can tailor agents to courses and integrate them into learning management systems, ensuring that support is accurate, relevant and aligned to course objectives (see [Case Study: Empowering Faculty Use of AI Agents for Student Success](#)).
 - York University, Canada launched a secure, scalable AI solution with a no-code, user-friendly interface that makes it accessible to all users. It enabled effortless integration of diverse knowledge sources into AI solutions to enhance system intelligence and ensure accurate, hallucination-free responses (see [Case Study: Scale University Adoption of Generative AI With a No-Code Platform](#)).
 - University of South Florida enabled Copilot for all users, created an M365 Copilot preview program and used MS Security Copilot to enhance IT security insights. It developed custom AI chatbots and AI-generated IT ticketing systems, while researchers are using high-performance computing resources, Microsoft Azure AI Hub and OpenAI Studio to develop AI applications and use fine-tuned LLMs for research (see [Case Study: Enhance AI Adoption and Operational Efficiency in Higher Education](#)).
- Encourage use-case-specific exploration of how to leverage AI positively in teaching, learning, research and administration (see [Generative AI Use-Case Assessment to Prioritize Education Investments](#)).

Support an agile curriculum development strategy and capability:

- Recognize that traditional curriculum development processes are often characterized by a lengthy cycle and inflexible structures. Work with faculty to review the current agility of your current curriculum development process and the extent to which curriculum development emphasizes continuous improvement, or allows for iterative changes based on real-time feedback from students, faculty and employers.
- Evaluate faculty needs and abilities to embed AI literacy across their curricula as part of this review. ¹⁷

- Plan to use AI tools to assist in curriculum mapping, predictive analytics of course demand and automating the generation of learning pathways to transform curriculum development from reactive to proactive and thus provide a competitive advantage to your institution (see [Market Guide for Higher Education Catalog and Curriculum Management Solutions](#)).
- Initiate agile curriculum development by forming cross-functional teams that include faculty, academic leadership, students and IT leadership.
- Evolve your ability to support any nontraditional, nondegree programs and adult upskilling by implementing technology solutions to easily administer these programs (see [Market Guide for Continuing Education and Workforce Development Solutions](#)).
- Track rapidly evolving global, national and state standards for digital microcredentials. Deploy technology solutions that enable your institutions to award, store, verify and manage academic digital credentials, aligning your credentialing strategy with your institutional goals for employability, student enrollment, mobility and lifelong learning (see [Market Guide for Academic Digital Credential Solutions in Education](#)).

Provide learning experiences that enable workplace-realistic learning:

- Evaluate the quality of experience provided by your current learning platforms for initial and continuing development across all modes of delivery.
- Embed simulations to stimulate critical thinking and workplace decision-making scenarios that complement traditional delivery, thus accelerating learning and improving student engagement
- Ensure that your digital transformation strategy continues to reshape the education landscape. Do this by combining AI with augmented reality (AR), virtual reality (VR) and mixed reality (MR) to evolve how curricula are designed, delivered and experienced by students (see [Case Study: Leveraging Immersive Technology to Empower Executive Education](#) and [Innovation Insight: Immersive Technology in Education](#)).
- Embed internships, career coaching and student support as critical components in managing student success. ¹⁸

Review, refine and continuously improve:

- Understand that AI shockwaves and higher education disruption will continue to impact institutional success in the near term.
- Work with your leadership team to ensure alignment of your target institutional position and evolving institutional AI and technology strategies. The propagation of these waves in the next year will demand institutional agility, adaptability and action.
- Design future shockwave sensors by embedding KPIs and AI governance structures that monitor and drive appropriate change.
- Ensure regular review and refinement of strategy and practice to drive cultural and institutional change. The sector is still in the early stages of reinvention.

Evidence

¹ [Measuring AI Ability to Complete Long Tasks](#), METR.

² [Trends in College Pricing](#), CollegeBoard.

³ [Is a College Degree Worth It in 2024?](#), Pew Research Center.

⁴ [Colleges Are Now Closing at a Pace of One a Week. What Happens to the Students?](#), Youth Today.

⁵ [Global Outbound Student Growth at Decades-Low Pace](#), British Council.

⁶ [U.K. Graduates Facing Worst Job Market Since 2018 Amid Rise of AI, Says Indeed](#), The Guardian.

[Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data](#)

[Canaries in the Coal Mine? Six Facts About the Recent Employment Effects of AI](#), Stanford.

⁷ [2025 EDUCAUSE AI Landscape Study: Into the Digital AI Divide](#), EDUCAUSE.

⁸ [America's Talent Strategy: Building the Workforce for the Golden Age](#), U.S. Department of Labor. [PM Launches National Skills Drive to Unlock Opportunities for Young People in Tech](#), Gov.UK.

⁹ [Credential Engine](#).

- ¹⁰ [Guidance for Generative AI in Education and Research](#), UNESCO.
 - ¹¹ [Re: Guidance on the Use of Federal Grant Funds to Improve Education Outcomes Using AI](#), U.S. Secretary of Education.
 - ¹² [Ethical Guidelines for Educators on Using AI](#), European Commission.
 - ¹³ [Critical Skills for the 21st Century](#), Florida International University. [Soft Skills: Critical Thinking](#), Kent State University.
 - ¹⁴ [Reimagining Higher Education: Seneca Polytechnic Set to Transform Postsecondary Learning With Integration of Microsoft Azure AI](#), Seneca Polytechnic.
 - ¹⁵ [Advancing the Responsible and Trustworthy Use of AI in Business and Society](#), Wharton.
 - ¹⁶ [Connecting and Innovation With Interactive AI](#), NUS.
 - ¹⁷ [Making AI a Core Literacy and Methodology](#), University of Florida.
 - ¹⁸ [Students' expectations and experiences of higher education](#), OfS.
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Recommended by the Author

Some documents may not be available as part of your current Gartner subscription.

[The Future of Higher Education – Vision 2035](#)

[Market Guide for Higher Education Catalog and Curriculum Management Solutions](#)

[Walking the Golden Path to Value: 2025 IT Symposium/Xpo Keynote Insights](#)

[Higher Education Ecosystem 2030: Planning in the Face of Radical Uncertainty](#)

[AI's Ripple Effect on Jobs and Organizations](#)

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